Machine learning Assignment 1 (Matlab)

Main Assignment Script

- Script: *main_assignment_script.m*
- Purpose: This is the main script that runs the entire program without user intervention. It reads preprocessed weather data, shuffles it, splits it into training and test datasets, and calls the Naive Bayes classifier to make predictions and calculate the error rate.

Task 1: Data preprocessing

- Script: *weather_data_processing.m*
- Objective: Preprocess the weather data.
  - Reads raw weather data from *weather_data.txt*.
  - Converts categorical weather attributes into integers.
  - Saves the processed data into a text file (*processed_weather_data.txt*), which is used in subsequent tasks.

Task 2: Build a naive Bayes classifier

- Script: *naive_bayes_classifier.m*
- Objective: Implement a Naive Bayes classifier without Laplace smoothing.
  - Reads the processed weather data for classification.
  - Calculates prior probabilities and likelihoods.
  - Uses a cell array to store likelihoods for each feature and class combination. This allows the model to handle different numbers of unique values for different features effectively.
  - Predicts the class (Yes/No) for test data.
  - Outputs predictions and the error rate.

Mathematical Background

Prior Probability
The prior probability for each class *Ci* is given by:

$$P(Ci) = \frac{Count(Ci)}{N}$$

Where:

- *Count (Ci)* is the number of instances in class *Ci*
- *N* is the total number of instances.

The vector **prior** in the MATLAB script contains the probabilities for each class.

Likelihood Calculation
The likelihood of observing feature *Xj* given class *Ci* is calculated as:

$$P(Xj|Ci) = \frac{Count(Xj\ in\ Ci)}{Count(Ci)}$$

Where *Count (Xj in Ci*) is the number of times feature *Xj* takes on a specific value within instance in *Ci*. The cell array **likelihoods {i, j}** contains the calculated likelihood for each unique value of feature *Xj* for class *Ci*.

Posterior Probability
The posterior probability for class *Ci* given feature vector *X* is given by:

$$P(Ci|X) = \frac{P(X|Ci) \cdot P(Ci)}{P(X)}$$

For classification, we do not need to compute *P(X)* since it is constant for all the classes:

$$P(Ci|X) \propto P(X|Ci) \cdot P(Ci)$$

So, the likelihood is calculated as:

$$P(X|Ci) = \prod_{j=1}^{m} P(Xj|Ci)$$

Where m is the number of features.
The posterior(j) saves the product of the prior and likelihood for each class in the script.
The posterior probabilities of all the classes are then compared. The highest posterior probability is taken as the prediction.

Output
The classifier outputs predictions based on the highest posterior probability:

$$Prediciton = argmaxP(Ci|X)$$

Task 3: Improve Classifier with Laplace Smoothing

- Script: *naive_bayes_classifier_laplace.m*
- Objective: Make the classifier robust to missing data with Laplace (additive) smoothing.
  - Reads the processed weather data for classification.
  - Determine the number of unique levels for each feature.
  - Calculates likelihoods using Laplace smoothing.
  - Calls the Naive Bayes classifier with Laplace smoothing function using numLevels calculated from the main script.

Mathematical Background

Likelihood Calculation with Laplace smoothing
The likelihood calculation with Laplace smoothing is given by:

$$P(x = i) = \frac{n_i + \alpha}{N + \alpha v}$$

Where:

- n_i is the count of the feature value *i* in class *Ci*.
- α is the Laplace smoothing parameter.
- *N* is the total count of instances in class *Ci*.
- v is the number of unique values for the feature.
- Outputs predictions and the error rate.